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Semantic Importance-Aware Communications With Semantic Correction Using Large Language Models

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The code can be available at <https://github.com/SSG2019/ULSC>.

ABSTRACT Semantic communications, a promising approach for agent-human and agent-agent interactions, typically operate at a feature level, lacking true semantic understanding. This paper explores understanding-level semantic communications (ULSC), transforming visual data into human-intelligible semantic content. We employ an image caption neural network (ICNN) to derive semantic representations from visual data, expressed as natural language descriptions. These are further refined using a pre-trained large language model (LLM) for importance quantification and semantic error correction. The subsequent semantic importance-aware communications (SIAC) aim to minimize semantic loss while respecting transmission delay constraints, exemplified through adaptive modulation and coding strategies. At the receiving end, LLM-based semantic error correction is utilized. If visual data recreation is desired, a pre-trained generative artificial intelligence (AI) model can regenerate it using the corrected descriptions. We assess semantic similarities between transmitted and recovered content, demonstrating ULSC's superior ability to convey semantic understanding compared to feature-level semantic communications (FLSC). ULSC's conversion of visual data to natural language facilitates various cognitive tasks, leveraging human knowledge bases. Additionally, this method enhances privacy, as neither original data nor features are directly transmitted.

INDEX TERMS Semantic communications, semantic error correction, large language models, generative artificial intelligence.

I. INTRODUCTION

RECENT developments in semantic communications, previously underexplored or overlooked, have propelled this field to the forefront of scholarly inquiry [1]. This renewed focus is driven by several key factors: advancements in deep learning technologies, the evolution of applications involving both agent-human and agent-agent communications, and a growing recognition of its critical role in transcending the limitations of system designs governed by Shannon's communication theory [2], [3], [4], [5].

A. RELATED WORK

The evolution of semantic communications has been marked by a transition from transmitting original data to channel-robust, low-dimensional latent space representations. This paper delineates three primary categories based on the nature of these representations.

1) CONTINUOUS LATENT SPACE REPRESENTATIONS

Continuous latent space representations involve encoding original data into continuous-valued vectors. Various

methodologies, including autoencoder (AE) [6], [7], variational autoencoder (VAE) [8], and generative adversarial network (GAN) [9], facilitate this process. These methods enable the encoding of source data into a compact continuous space, emphasizing essential features while omitting redundant elements, thereby optimizing transmission efficiency. This approach aids in tasks such as retrieval, classification, and reconstruction, as exemplified in studies focused on image reconstruction [10] and image retrieval [11], which demonstrate enhanced performance compared to traditional communication systems. Recently, semantic communication applications have expanded to encompass text [12], [13], audio [14], [15], video [16], and graph data [17], showcasing promising results. However, integration with existing digital communication systems remains a challenge due to compatibility issues, necessitating substantial system-wide updates [18].

2) DISCRETE LATENT SPACE REPRESENTATIONS

This category involves transforming continuous semantic vectors into discrete codewords, aligning with digital communication systems' structures. Xie and Qin [19] developed quantization-based feasible constellations for existing systems. The VAE's Gaussian latent variable distribution was modified to a Bernoulli distribution for discrete representation acquisition [20], although this added complexity to the loss computation. Nemati et al. [21] introduced a vector quantized VAE (VQ-VAE) based joint source and channel coding approach, employing a shared codebook between transmitter and receiver for mapping semantic vectors to codewords. This method effectively clusters semantic vectors, with codewords serving as class centers. Subsequent studies [22], [23], [24], [25] have proposed variations in encoder and decoder architectures, while retaining the discrete representation principle through codebook construction.

3) NATURAL LANGUAGE REPRESENTATIONS

Compared to feature-level systems, this approach bridges the semantic gap between low-level data and high-level human-related meanings. It uses a natural language representation to map data into semantic domains that can be understood by humans. This approach has been explored in a number of works. For example, [26] studies how to use deep learning techniques to transform images into informative text descriptions. Reference [27] further explores how this conversion process can be optimized to produce a more accurate text representation. Focusing on the wireless communication environment, [28] studies how to effectively transmit these semantically rich text descriptions. This approach not only improves the quality of interaction between machines and humans, but also enhances the semantic communication ability between machines.

B. CONTRIBUTIONS OF THIS WORK

This paper introduces a novel understanding-level semantic communications (ULSC) system, adeptly designed for seamless integration with existing communication networks. The

system is characterized by its utilization of advanced components such as pre-trained image caption neural networks (ICNNs), and large language models (LLMs). Compared with existing mixed image-text transmission systems [26], [27], [28], our proposed ULSC not only focuses on feature extraction and encoding, but also prioritizes processing and transmission of the most critical semantic information by introducing a semantic importance aware mechanism. The contributions of this work are outlined as follows:

- This work introduces a novel method for quantifying the semantic importance of transmitted messages using pre-trained LLMs. Unlike existing methods that focus on feature-level transmission, our approach enables frame-level prioritization of semantically critical information, optimizing bandwidth usage and transmission reliability in challenging network conditions.
- We present a semantic error correction mechanism that leverages the deep contextual understanding of LLMs (e.g., Bidirectional Encoder Representations from Transformers (BERT)). By predicting and filling in missing or corrupted frames of transmitted messages, our approach significantly reduces semantic loss.
- Our framework enhances privacy protection by transforming raw visual data into high-level semantic representations, which are more difficult to reverse-engineer. This provides a higher level of security compared to traditional feature-level communication systems, as unauthorized entities cannot easily infer sensitive information from intercepted transmissions. We demonstrate the effectiveness of this approach through experiments involving model inversion attacks.
- We provide a thorough evaluation of our proposed ULSC system, comparing it against traditional feature-level semantic communication methods. Our results demonstrate that the use of LLMs not only reduces semantic loss but also improves system robustness and privacy protection.

C. ORGANIZATION

The remainder of this paper is methodically structured to guide readers through our comprehensive research in ULSC. Section II begins with an in-depth examination of the ULSC system model, detailing the transceiver structures and mathematical modeling essential to our approach. This is followed by Section III, which delves into the transmitter functions, particularly focusing on the conversion of visual data to natural language and the quantification of frame importance. Section IV discusses the implementation of frame-level, semantic importance-aware communications over networks, incorporating a thorough performance evaluation and system optimization in the context of frame erasure channels. In Section V, the paper shifts focus to the receiver's role, elaborating on the use of a pre-trained language model for semantic error correction and the application of pre-trained generative AI models for visual data generation. Section VI

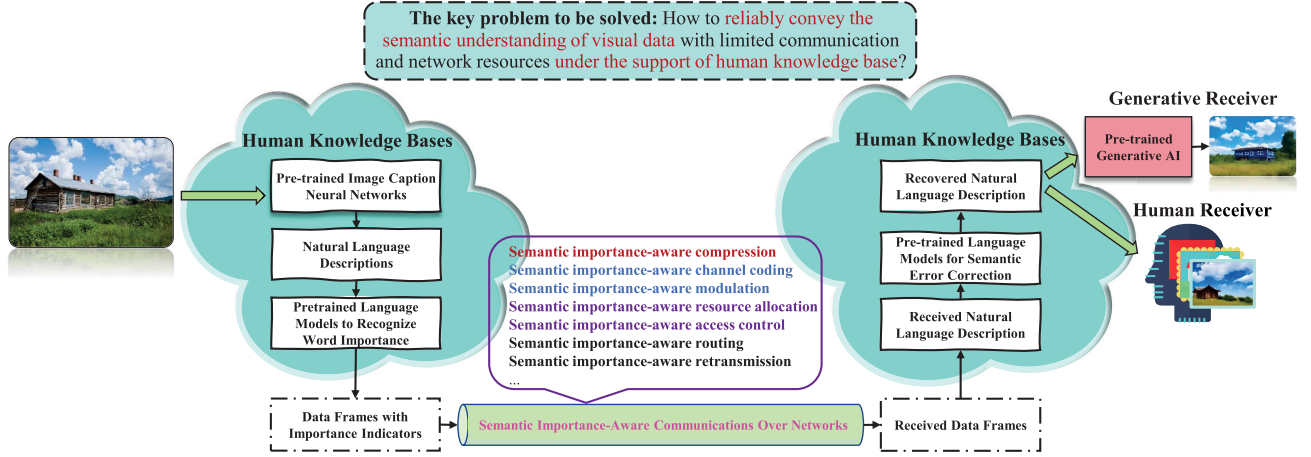


FIGURE 1. Understanding-level semantic communications to reliably transmit the semantic understanding of visual data with limited communication between the transmitter and the receiver under the support of human knowledge bases.

presents our experimental studies and discussions, offering crucial insights into the practical applications and effectiveness of the proposed system. In Section VII, some key insights and future research directions are discussed. Finally, the paper culminates in Section VIII with conclusions that summarize our key findings and contributions.

II. SYSTEM MODEL AND PROBLEM DESCRIPTION

We consider an understanding-level semantic communication system, as shown in Fig. 1. The system aims to convey the semantic essence of visual information in the context of limited communication bandwidth and network resources. This is achieved by transforming visual data into natural language descriptions and employing semantic error correction techniques.

A. SYSTEM MODEL

The ULSC system model includes a transmitter with a visual input function that processes visual data through an image captioning neural network (ICNN) and generates natural language descriptions. Let $\mathbf{S} \in \mathbb{R}^N$ represent the input visual data, where N is the dimension of the image data. The ICNN processes this data and converts it into a natural language description $\mathbf{X} \in \mathbb{Z}^D$, where $\mathbb{Z}$ is the vocabulary and D is the number of words. The mathematical formula for this conversion is:

$$\mathbf{S} \in \mathbb{R}^n \rightarrow \mathbf{X} \in \mathbb{Z}^d$$

$$\mathbf{X} = f_{\alpha}(\mathbf{S}), \quad (1)$$

where $f_{\alpha}(\cdot)$ is the ICNN function, parameterized by the neural network weight α .

For transmission, the natural language description $\mathbf{X}$ is divided into L frames, each of which is encoded as a binary sequence by source coding and channel coding. Let $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_L\}$ represent the encoded binary sequence of L frames. These frames are then transmitted over the network. It is worth mentioning that in order to optimize transmission, we prioritize semantically important information.

During transmission, the encoded frames $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_L\}$ may have errors or be lost. The receiver uses a semantic error correction mechanism based on a large language model (LLM) to recover the lost frames, ensuring that the transmitted semantic information is preserved. The mathematical expression of semantic error correction is:

$$\tilde{\mathbf{X}} = f_{\theta}(\hat{\mathbf{X}}), \quad (2)$$

where $\hat{\mathbf{X}}$ and $\tilde{\mathbf{X}}$ are the natural language representation after transmission and the natural language representation after semantic error correction, respectively. $f_{\theta}(\cdot)$ is the semantic error correction function, parameterized by θ .

After obtaining the corrected natural language description $\tilde{\mathbf{X}}$, it will be passed to the final receiver. The receiver may be a visually impaired individual who receives information through acoustic channels. Alternatively, the receiver could be a generative AI model with natural language representation as text prompt. This process can be formalized as:

$$\hat{\mathbf{S}} = f_{\beta}(\tilde{\mathbf{X}}), \quad (3)$$

where $\hat{\mathbf{S}}$ represents the generated speech, image, or other formats of content. $f_{\beta}(\cdot)$ denotes speech generation model, generative AI model, or some other model, and β represents its parameters.

B. PROBLEM DESCRIPTION

In ULSC system, raw visual data is first transformed into a series of natural language descriptions. These descriptions are then divided into multiple data frames for encoding and preparation for transmission. However, in the process of network transmission, these encoded frames may be wrong or lost due to various factors such as noise interference and signal attenuation. To address this challenge, we employ semantic error correction techniques on the receiving end to recover lost or damaged frames as much as possible, thereby maximizing the retention of transmitted semantic information.

The goal of this system is to minimize the overall semantic information loss during transmission. This means that we not only focus on the integrity of each data frame, we also want to optimize the delivery of critical information with limited resources. In subsequent chapters, we will detail how to implement these theories in practical applications and verify the effectiveness of these strategies through experiments. This includes data preprocessing technology on the sending side, semantic communication strategy in the network, and semantic recovery mechanism on the receiving side.

III. TEXTUAL ENCODER AND FRAME IMPORTANCE QUANTIFICATION

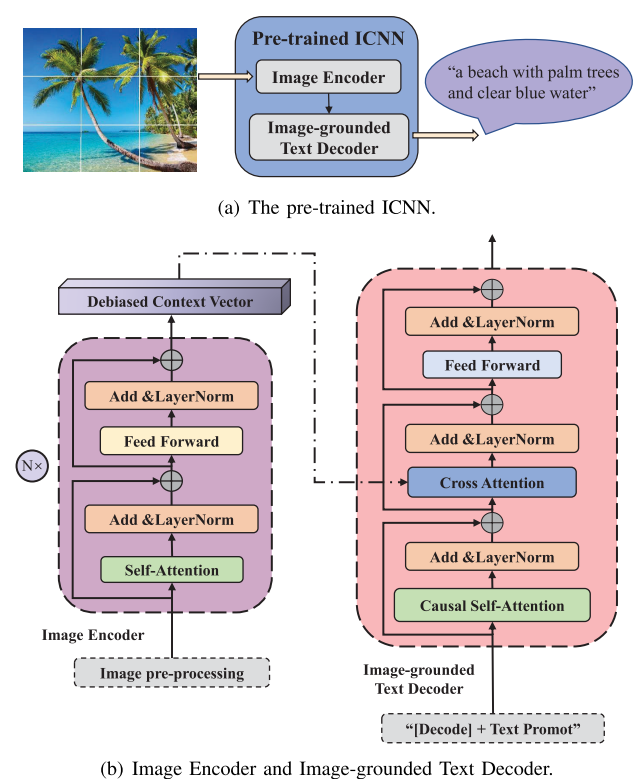
This section delves into the intricacies of the textual encoder within the visual data-to-text conversion process and the methodology for quantifying frame importance, which are fundamental components of the ULSC transmitters.

A. VISUAL DATA TO NATURAL LANGUAGE CONVERSION

The primary objective of ULSC is to facilitate the comprehension of visual data for visually impaired individuals or generative AI systems by utilizing expansive human knowledge bases. The transformation of visual data into natural language is a critical and intricate process within ULSC, encompassing aspects of computer vision (CV) and natural language processing (NLP). This complex process, referred to as image captioning in the field of deep learning, requires a deep understanding of visual content to generate accurate and meaningful descriptions.

Significant advancements have been made in addressing this challenge. Notable approaches include the encoder-decoder model, as demonstrated in works by Vinyals et al. [29] and Karpathy et al. [30]. These models employ convolutional neural networks (CNNs) to extract image features and recurrent neural networks (RNNs) or long short-term memory (LSTM) networks to generate corresponding textual descriptions. To enhance the accuracy and relevance of these descriptions, an attention mechanism has been integrated into the encoder-decoder framework. This mechanism allows the model to focus on different segments of the image during the generation process, as detailed in the studies by Xu et al. [31] and Li et al. [32].

Building on these advancements, our research has adopted the pre-trained ICNN proposed by Li et al. [32]. Fig. 2(a) shows the workflow of the ICNN. The input image is divided into a number of small blocks, which are converted into a high-dimensional feature representation after passing through the image encoder. These image feature vectors are then passed to an image-based text decoder, which generates a text description related to the image content. Fig. 2(b) shows the structure of the image encoder and image-grounded text decoder in detail. The image encoder consists of N Transformer layers. Each Transformer layer includes a Self-Attention layer, Feed Forward network, and LayerNorm. These layers extract image features by Self-Attention mechanism, and further process them by Feed



(b) Image Encoder and Image-grounded Text Decoder.
FIGURE 2. The adopted pre-trained ICNN and its details.

Forward network, and finally generate image feature vectors. In image-grounded text decoder, each Transformer layer also contains LayerNorm, Feed Forward network, and Causal Self-Attention layer. In addition, a Cross Attention layer is added, which is used to combine image features and text information to generate text descriptions related to the image content.

Notably, the generated text's length is adjusted according to the complexity of the image. For simpler images, the text generated will be shorter, while for more complex images with richer semantic content, the text length increases accordingly. This dynamic text generation mechanism ensures that the description is neither too long nor too short. The image encoder also analyzes the image's complexity by evaluating various aspects such as the number of objects, scene diversity, and the relationships among visual elements. The encoder generates feature vectors that inform the text decoder about these aspects, enabling the system to prioritize semantically important elements in the image. For example, when processing an image with multiple objects or scenes, the model identifies and emphasizes key objects and their interactions, and generates text that reflects these important elements. Further details of the model and its implementation can be found in [32].

In traditional communication systems, there is a risk of visual privacy disclosure when transmitting raw images or low-level features. An attacker can use methods such as model inversion [7] to reconstruct the original image (e.g., facial image), and then obtain the user's skin color

TABLE 1. The importance of each word in example sentences with and without semantic error correction.

Example 1	Word	A	beach	with	palm	trees	and	clear	blue	water
	Semantic Error Correction	0	0.0386	0	0	0	0	0	0	0.0457
	No Semantic Error Correction [18]	0.0197	0.0515	0.0389	0.0131	0.0253	0.0357	0.0109	0.0145	0.0342
Example 2	Word	A	bald	eagle	flying	over	a	body	of	water
	Semantic Error Correction	0	0.0092	0.0780	0.0593	0	0	0	0	0.1045
	No Semantic Error Correction [18]	0.0227	0.0130	0.0758	0.0474	0.0229	0.0438	0.0606	0.0399	0.0844
Example 3	Word	A	forest	filled	with	lots	of	tall	trees	—
	Semantic Error Correction	0	0.0364	0.0305	0	0.0432	0	0.0104	0.0740	—
	No Semantic Error Correction [18]	0.0426	0.0423	0.0145	0.0244	0.0574	0.0219	0.0132	0.0477	—

information, or directly use it for facial recognition. Our approach converts visual information into natural language descriptions. This description preserves only the most critical information in the image for semantic understanding, filtering out details that are not relevant to the original pixel-level features. This transformation increases the difficulty of countering model inversion attacks because there are significant differences in feature space between the natural language description and the original image, reducing the likelihood of the attacker recovering the original visual information.

B. FRAME IMPORTANCE QUANTIFICATION

The concept of frame importance is pivotal in semantic importance-aware communications. It pertains to the relevance and significance of data within a frame for the comprehensive understanding of the overall context. Due to its content-dependent and dynamic nature, the quantification of semantic importance is intricate and often requires deep contextual understanding and domain-specific knowledge. Establishing universal metrics for this quantification is challenging, as it may vary across different contexts.

In our approach, we amalgamate human judgment with automated methods, including natural language processing techniques, machine learning algorithms, and user feedback, to quantify the semantic importance of data within frames. This multifaceted approach is a subject of ongoing research. Specifically, we utilize pre-trained language models, such as ChatGPT [33] and BERT [34], to measure the importance of words and frames based on the semantic variations resulting from their erasure, as detailed in [18].

Our method involves comparing the original message, denoted as $\mathbf{m}_0$, with a modified version, $\mathbf{m}_l$, where the l th frame is absent but replaced with a completion frame generated by a LLM. The semantic importance of the l th frame is then quantified as:

$$F_l = 1 - \phi(\mathbf{m}_0, \mathbf{m}_l), \quad (4)$$

where $\phi(\mathbf{m}_0, \mathbf{m}_l) = \frac{B_\psi(\mathbf{m}_0)^T B_\psi(\mathbf{m}_l)}{\|B_\psi(\mathbf{m}_0)\| \cdot \|B_\psi(\mathbf{m}_l)\|}$ measures the semantic similarity between $\mathbf{m}_0$ and $\mathbf{m}_l$. This similarity is calculated using the cosine similarity of the embeddings generated by a pre-trained BERT model, $B_\psi(\cdot)$, for both messages.

236 that cosine similarity of Embedding, which is generated by pre-trained BERT model, $B(\cdot)$ for both msg, original and corrected by LLM

This approach ensures that the quantification of importance reflects the ability of semantic error correction mechanisms to recover or predict missing information.

Notably, this method allows the transmitter to independently assess the importance of words or frames without requiring interaction with the receiver. By simulating a semantic error corrector, the transmitter can determine the potential for each word or frame to be accurately reconstructed or predicted by the receiver's error correction system.

Table 1 presents a comparative analysis between our proposed method, which includes semantic error correction, and the previously discussed method that does not incorporate this aspect (as detailed in [18]). A striking observation from this comparison is the prevalence of zero importance scores for most words in the context of semantic error correction. This outcome indicates that many words, when lost, can be effectively predicted and completed by the receiver's error correction mechanisms. Consequently, the ULSC framework can prioritize transmitting words or frames that are less likely to be accurately reconstructed if lost, thereby enhancing communication efficiency and reliability.

This approach represents a significant advancement in the field of semantic communications, particularly in applications involving ULSC, as it provides a more nuanced understanding of data importance and optimizes data transmission based on the receiver's error correction capabilities.

IV. SEMANTIC IMPORTANCE AWARE COMMUNICATIONS OVER NETWORKS

The concept of semantic importance-aware communications centers around the prioritization of semantically significant frames during transmission. This approach is instrumental in optimizing data traffic management and ensuring that vital information is accorded the utmost priority. Such prioritization is crucial in minimizing delays and augmenting the overall efficiency of the system. In this framework, frames of greater semantic importance are transmitted or processed in precedence to those of lesser significance.

Semantic importance-aware communications exhibit versatility across multiple layers. At the foundational physical layer, diverse modulation and coding schemes are employed

Semantic Importance Formula using original and modified msg,

SIAC work on various layer VOLUME 3, 2025 foundational Physical layer, diverse modulation and coding schemes use and apply on trans. data to define various level of importance

to transmit data of varying levels of importance. For example, data of higher importance might be encoded using more robust error-correcting codes or transmitted using lower-order modulation to enhance reliability.

In specific physical layer technologies, such as cellular systems, there exists the capability for dynamic allocation of channels to frames based on their importance. Frames of higher importance may be allocated to channels experiencing less congestion. This strategy, situated at the physical layer, encompasses the retransmission of data in instances of errors or collisions during the initial transmission. In scenarios where significant data encounters errors, it may be retransmitted with elevated priority to assure a more reliable delivery.

Furthermore, power control mechanisms within wireless communication can allocate preferential treatment to certain frames by modulating their transmit power. This adjustment ensures that frames of higher importance receive better signal quality and encounter reduced interference.

It is imperative to recognize that while some of these methodologies are implemented directly within the physical layer, many communication systems integrate both physical layer and higher-layer techniques, such as importance-aware routing and access control, to realize efficient and reliable semantic communication.

To elucidate semantic importance-aware communication and its integration with existing communication frameworks, we examine the semantic importance-aware adaptive modulation and coding selection (MCS) as a representative example. In this model, transmitted messages are divided into a total of L frames, each capable of carrying a payload of A_l bits, inclusive of both header and data ($\mathbf{b}_l$). This semantic communication system deviates from traditional systems that predominantly focus on the accuracy of transmitted bits or symbols. Instead, it emphasizes the paramount objective of enabling the receiver to accurately comprehend and extract the intended information relayed by the transmitter. In this context, traditional metrics such as bit/symbol/frame error or loss probability are insufficient as evaluation criteria. Instead, we advocate for the adoption of semantic loss as a metric to quantify the integrity of the information exchange between the transmitter and receiver.

The expected semantic loss (SL) incurred by a frame loss is determined by the semantic importance of the messages contained within each frame (F_l), combined with the frame loss probabilities ($P_{l,frame}^{Loss}$). It is essential to acknowledge that, in the sphere of semantic communications, each frame conveys unique messages, imparting varying degrees of semantic significance. This variation in importance amongst frames, often overlooked in conventional data-oriented communication systems, is of paramount importance in semantic communications. Notably, the omission or inaccuracy of messages bearing critical information can lead to a disproportionately high semantic loss. Therefore, the expected semantic loss due to frame loss is characterized by the integration of the semantic importance of the messages and their

corresponding loss probabilities as

$$SL = \sum_{l=1}^L F_l P_{l,frame}^{Loss} \quad \text{semantic loss} \quad (5)$$

The frame loss probability hinges upon the MCS and the channel status. Let $P_{l,frame}^{Correct}$ denote the successful reception and recovery of the l th frame and we have

$$P_{l,frame}^{Loss} = 1 - P_{l,frame}^{Correct} \quad \text{frame loss prob.} \quad (6)$$

The correct probability can be derived as

$$P_{l,frame}^{Correct} = (P_{l,block}^{Correct})^{N_B} \quad \text{correction probability} \quad (7)$$

where $P_{l,block}^{Correct}$ is the correct probability of coding block. $N_B = \lceil A_l / (R_l B_l) \rceil$ is the number of coding blocks per frame, where R_l is the coding rate, B_l is the coding block length.

$$P_{l,block}^{Correct} = \sum_{i=0}^{C_l} \binom{B_l}{i} P_{l,bit}^{Error}(\mathcal{M}_l, \gamma)^i \left(1 - P_{l,bit}^{Error}(\mathcal{M}_l, \gamma)\right)^{B_l-i}, \quad (8)$$

where C_l denotes the number of corrigible errors and it is determined by the choosing coding scheme. In general, $C_l = \lfloor \frac{d_{l,min}-1}{2} \rfloor$, where $d_{l,min}$ represents the minimum number of bit positions by which any two distinct codewords differ. $P_{l,bit}^{Error}(\mathcal{M}_l, \gamma)$ is the bit error probability determined by the symbol error probability $P_{l,sym}^{Error}(\mathcal{M}_l, \gamma)$, and the adopted bit-to-symbol mapping scheme. It can be approximated accurately as $P_{l,bit}^{Error}(\mathcal{M}_l, \gamma) \simeq \frac{P_{l,sym}^{Error}(\mathcal{M}_l, \gamma)}{\log_2 M_l}$ in high SNR region in case of Gray mapping, where M_l denotes the modulation order for the adopted $\mathcal{M}_l$ modulation scheme. $\gamma = |h|^2 E_b R_l \log M_l / N_0$ represents the SNR, where h is the channel coefficient, E_b stands for power per bit, and N_0 denotes the noise power. The SEP of typical modulation schemes has been summarized in Table 2.

Therefore, based on the derived symbol error probability, we can obtain the expression of the frame loss probability as

$$P_{l,frame}^{Loss} = 1 - \left[\sum_{i=0}^{C_l} \binom{A_l}{i} \left(\frac{P_{l,sym}(\mathcal{M}_l, \gamma)}{\log_2 M_l} \right)^i \times \left(1 - \frac{P_{l,sym}(\mathcal{M}_l, \gamma)}{\log_2 M_l} \right)^{A_l-i} \right]^{N_B}. \quad (9)$$

Substituting (4) and (9) into (5) yields

$$SL = \sum_{l=1}^L \left(1 - \frac{B_\psi(\mathbf{m}_{l,t})^T B_\psi(\mathbf{m}_{l,r})}{\|B_\psi(\mathbf{m}_{l,t})\| \cdot \|B_\psi(\mathbf{m}_{l,r})\|} \right) \left\{ 1 - \left[\sum_{i=0}^{C_l} \binom{A_l}{i} \left(\frac{P_{l,sym}(\mathcal{M}_l, \gamma)}{\log_2 M_l} \right)^i \left(1 - \frac{P_{l,sym}(\mathcal{M}_l, \gamma)}{\log_2 M_l} \right)^{A_l-i} \right]^{N_B} \right\}. \quad (10)$$

To ensure that the semantic communication system meets its latency requirements, it is imperative to assess the overall

TABLE 2. SEP of various modulation schemes.

Schemes	$P_{l,sym}(\mathcal{M}_l, \gamma)$
M -PSK	$Q(\sqrt{2\gamma}) + \frac{2}{\sqrt{\pi}} \int_0^\infty \exp[-(u - \sqrt{\gamma})^2] Q\left(\sqrt{2}u \tan \frac{\pi}{M}\right) du$
M -QAM	$4 \left(\frac{\sqrt{M_l}-1}{\sqrt{M_l}} \right) Q\left(\sqrt{\frac{3\gamma}{(M_l-1)}}\right) - 4 \left(\frac{\sqrt{M_l}-1}{\sqrt{M_l}} \right)^2 Q^2\left(\sqrt{\frac{3\gamma}{(M_l-1)}}\right)$

transmission time for the conveyed messages. Assuming a transmission rate of R_s in symbols/seconds for the semantic system under consideration, the transmission time for the l -th frame can be determined as

$$T_l = \frac{A_l}{\log_2(M_l) R_l R_s}. \quad (11)$$

Consequently, the total transmission delay can be evaluated as

$$T = \sum_{l=1}^L T_l = \sum_{l=1}^L \frac{A_l}{\log_2(M_l) R_l R_s}. \quad (12)$$

Let $\mathcal{M}_l$ be the modulation scheme chosen from a modulation candidate $\mathcal{M}_{all}$ for the transmission of the l th frame. Let $\mathcal{C}_l$ be the coding scheme chosen from a coding candidate $\mathcal{C}_{all}$ for the transmission of the l th frame. The modulation scheme $\mathcal{M}_l$ will determine the symbol error rate (i.e., $P_{l,sym}(\mathcal{M}_l)$) and the number of bits per symbol (i.e., $\log_2(M_l)$). The coding scheme $\mathcal{C}_l$ will determine error-correcting capabilities (i.e., C_l) and code rates (i.e., R_l). Both of them are key factors that dominate the semantic loss and transmission delay. Aiming at minimizing the semantic loss within a transmission delay constraint, we formulate the semantic-aware adaptive modulation and coding problem as

$$\begin{aligned} \min \quad & SL \\ \text{s.t.} \quad & T \leq T_{th}, \\ & \mathcal{M}_l \in \mathcal{M}_{all}, \\ & \mathcal{C}_l \in \mathcal{C}_{all}. \end{aligned} \quad (13)$$

The problem delineated in (13) is inherently a discrete optimization problem. To determine the optimal solution, one could theoretically conduct an exhaustive search of all possible solutions. This approach, however, escalates in computational complexity exponentially with the number of frames, following the formula $\mathcal{O}(|\mathcal{M}_{all}|^L |\mathcal{C}_{all}|^L C_1)$, where C_1 signifies the complexity of computing SL and T for each modulation and coding combination.

In order to alleviate the computational burden of this method, we propose a semantic-aware greedy search method. We randomly generate many feasible modulation and coding candidates and output the one with minimum semantic loss and acceptable transmission delay below the threshold. By applying such greedy search method, the computational complexity is significantly reduced to $\mathcal{O}(N_{set} LC_1)$, where N_{set} is the number of candidates the algorithm searches. This method provides a more computationally efficient approach, striking a balance between achieving near-optimal solutions

and adhering to constraints of semantic importance and transmission delay.

V. SEMANTIC CORRECTION AND VISUAL DATA GENERATION

The process of receiving semantically important signal representations, which are detected as bits, involves error correction for rectifiable errors, while uncorrectable frames are discarded. This study proposes a novel approach that leverages a LLM for word completion in scenarios where frames are lost.

A. SEMANTIC CORRECTION METHODOLOGY

Despite efforts to prioritize the transmission of semantically significant frames and quantify their importance, the reliability of information transmission cannot be guaranteed due to unpredictable factors such as wireless channel fading and noise interference. These factors can distort or result in the loss of information during transmission. In the ULSC process, the intermediate state is the natural language description. If this is not transmitted reliably, it can hinder the receiver's understanding and decision-making. This issue is particularly critical in applications involving visually impaired individuals, where misunderstandings or delays could compromise personal safety. Therefore, addressing the challenge of completing missing frames in received natural language descriptions is crucial for ensuring reliable semantic communication.

To tackle this challenge, this study proposes the use of a pre-trained LLM, such as BERT, which leverages contextual information and pre-training knowledge to infer and recover missing frames. The specific steps of this process are outlined as follows. First, the input sentences are preprocessed to fit the BERT model. Suppose we have a sentence: "A beach with palm [MASK] and clear blue water.", where [MASK] indicates the missing word. Using BERT's tokenizer, the sentence is tokenized and special tokens [CLS] and [SEP] are added to denote the start and end of the sentence, respectively. Next, the segmented sentence is converted into the corresponding ID in the vocabulary, which is used as input to the model. In addition, to accommodate the input format of BERT, these IDs need to be converted into tensor form. The preprocessed input data is fed into the BERT model. The BERT model consists of multiple Transformer layers, each of which is processed by Self-Attention mechanisms and Feed Forward Network. Among them, the mathematical expression of the Self-Attention mechanism is:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (14)$$

Q , K and V are query, key and value matrices respectively, and d_k is the dimension of the key vector. In BERT, each word representation interacts with other word representations through this mechanism to capture global information in the sentence. After passing the last layer of BERT, a contextual representation of each word can be obtained. Suppose the output is represented as a matrix Π with the shape (L_Π, D_Π) , where L_Π is the length of the sentence and D_Π is the dimension of the representation. It can be expressed as:

$$\Pi = [\pi_{[CLS]}, \pi_{[A]}, \dots, \pi_{[MASK]}, \dots, \pi_{[water]}, \pi_{[SEP]}]. \quad (15)$$

Next, the representation at position is extracted, and it is transformed into a probability distribution over the vocabulary through a fully connected layer and a softmax layer:

$$P(w|context) = \text{softmax}(\pi_{[MASK]}W + b), \quad (16)$$

where W and b are trainable weights and bias parameters, and $P(w|context)$ is the probability distribution of each word in the vocabulary. Finally, the word corresponding to the maximum value in the probability distribution is selected as the prediction result:

$$\hat{w} = \text{argmax}_w P(w|context). \quad (17)$$

By aligning the correction mechanism with human cognitive processes, this approach ensures a more reliable transmission of semantic information, thereby enhancing the overall effectiveness of communication systems, especially in critical applications.

B. VISUAL DATA GENERATION

Visual data generation is a pivotal component in enabling smart agents to interpret semantics and contextual nuances in received text. This process is achieved through sophisticated neural network models that transform textual descriptions into corresponding visual representations, such as images. Achieving a harmonious and logical alignment between the generated images and their textual counterparts is a challenging endeavor due to the inherent ambiguity in correlating images with natural language expressions and the necessity to maintain a high level of image realism.

Prominent developments in this field include OpenAI's DALL-E model [35] and Google's Imagen model [36], both designed for text-to-image generation. A notable limitation in these models is their computational efficiency, primarily because they function in pixel space. To manage computational constraints, these models initially produce images of 64×64 resolution, which are subsequently upsampled to 256×256 and finally to 1024×1024 , utilizing super-resolution techniques. A significant advancement is presented in [37], which introduced the Stable Diffusion model. The core idea is to start from the pure noise image and gradually recover to the target image through a series of reverse diffusion steps. Its operation in latent space offers enhanced computational

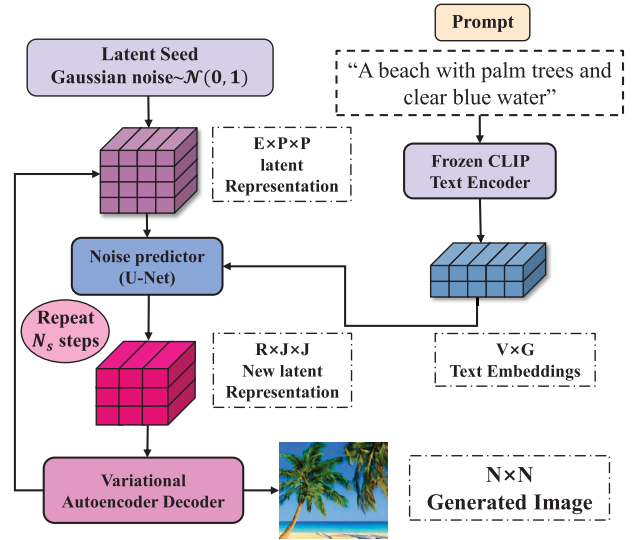


FIGURE 3. Visual data generation.

efficiency, making it a superior choice for integration into the ULSC system.

The image generation process of the Stable Diffusion model is illustrated in Fig.3. First, the user inputs a text prompt, such as “a beach with palm trees and clear blue water”. This text prompt is processed through a pre-trained CLIP text encoder, generating the corresponding text embeddings. The CLIP text encoder converts the text into a multi-dimensional vector that effectively captures the semantic information of the text. Meanwhile, an initial noise seed is sampled from a standard Gaussian distribution. These noise seeds serve as the initial input for the latent representation. The noise seeds are represented as a multi-dimensional matrix of a specific size, forming the starting point of the image generation process. Next, the noise seeds and text embeddings are fed into the noise predictor, which uses a U-Net model [38]. The U-Net model is a convolutional neural network capable of denoising while preserving crucial details in the image. At each timestep, the U-Net model predicts and removes noise based on the current latent representation and text embeddings, gradually generating a new latent representation. This process involves multiple iterative steps, with each step progressively removing some noise, transforming the latent representation from pure noise to a form closer to the target image. Throughout the denoising process, the text embeddings provide guidance for image generation, ensuring that the generated image aligns with the input text prompt. After completing all the denoising steps, the final latent representation is decoded through a VAE decoder, producing the final high-quality image. The VAE decoder converts the denoised latent representation into a complete image, significantly enhancing the detail and quality of the image, which matches the initial text prompt. Through this process, the Stable Diffusion model can start from random noise, incorporate the input text prompt, and gradually generate a high-quality image that corresponds to the description.

This process fully leverages the powerful features of pre-trained models, particularly the CLIP text encoder and the U-Net denoising network, ultimately achieving an efficient transformation from text to image.

VI. EXPERIMENTS AND DISCUSSIONS

In this section, we will demonstrate the advantage of semantic importance-aware communications through experiments and that of using LLM to assist semantic error correction. In particular, adaptive modulation and coding that is compatible with existing communication systems. Besides, we also demonstrate the advantage of ULSC compared to feature-based semantic communication in achieving high semantic similarity and protecting privacy.

A. EXPERIMENTAL SETUP

1) DATASETS AND PRE-TRAINED MODELS

We collected a comprehensive dataset comprising 1000 images sourced from the Internet, aimed at validating the effectiveness of our newly proposed method. This dataset comprises landscapes, animals, people, cartoons, and various other image categories. The resolution of each image is 256×256 . Additionally, in our experiments, we select two deep learning-based algorithms for comparison. These algorithms require extensive training on large-scale dataset. To ensure excellent algorithm performance, we train these algorithms using the ImageNet dataset [39]. ImageNet is a large-scale image dataset covering over 1,000 categories, ranging from animals to household items and various other categories. These images are not only diverse but also include objects from different angles and in various environments, making it an ideal dataset for training deep learning models. In our experiments, we adjust the resolution of each ImageNet image to 256×256 .

For our experiments, we utilize pre-trained models for different components of the ULSC system. Image caption neural networks for converting visual data into natural language descriptions are based on a model developed [32]. To evaluate the importance of semantic frames, we adopt BERT as our language model, specifically using the bert-base-uncased version from the Hugging Face Transformers library. The semantic error correction module uses the same BERT model. When visual data needs to be regenerated from natural language descriptions, we use a Stable Diffusion model. All the relevant code and detailed documentation for invoking the pre-trained model can be found on GitHub.¹

2) BASELINES

To verify the effectiveness of our proposed system, we compare it with two semantic communication schemes, including DeepJSCC [10] and SCDA [40].

- **DeepJSCC:** In DeepJSCC, CNNs are used to encode source images into a compact representation that is transmitted directly over a communication channel. The

receiver uses another CNNs to decode the received signal back to the original image.

- **SCDA:** SCDA is a new semantic communication system based on deep learning, which can process dynamic datasets without knowing the specific task of the receiver.

To more fully demonstrate the robustness and adaptability of our proposed system, we consider two channels: the frame erase channel and the AWGN channel. In particular, in the frame erase channel test, we consider two different data packaging strategies of DeepJSCC and SCDA systems when transmitting features. The first strategy is to package all features into 7 frames, and the other strategy is to package them into 14 frames. This is because the natural language generated from images in the proposed method contains a maximum of 14 words and a minimum of 7 words.

3) METRICS

In this paper, we focus on semantic similarities between transmitted and recovered semantic understanding. The commonly used peak signal-to-noise ratio (PSNR) mainly measures the difference of images at the pixel level, focusing on the average error between the original image and the reconstructed image. An image that scores high on the PSNR may not necessarily be visually closer to the original image. This is mentioned “semantic gap”, namely the image pixel level is similar, but there were significant differences in content and semantics. In this paper, we propose to calculate the image similarity between the original image $\mathbf{S}$ and the generated image $\hat{\mathbf{S}}$ to evaluate the performance of the algorithm.

$$\text{Image Similarity}(\mathbf{S}, \hat{\mathbf{S}}) = \frac{\mathbf{C}_\Phi(\mathbf{S}) \cdot \mathbf{C}_\Phi(\hat{\mathbf{S}})}{\|\mathbf{C}_\Phi(\mathbf{S})\| \|\mathbf{C}_\Phi(\hat{\mathbf{S}})\|}, \quad (18)$$

where $\mathbf{C}_\Phi(\mathbf{S})$ represents the embedding vector of the image $\mathbf{S}$ obtained through the pre-trained CLIP model $\mathbf{C}_\Phi$. The dimensionality of $\mathbf{C}_\Phi(\mathbf{S})$ depends on the size of the embedding layer in the CLIP model, which is 512 in this paper. Each component of the embedding vector is a real number ranging from $[-1, 1]$. The image similarity defined in (18) is a value between 0 and 1, indicating the degree of semantic similarity between the generated image and the original image, where 1 indicates the highest similarity and 0 indicates no similarity.

Besides, we focus on addressing a significant privacy concern in machine learning: model inversion attack. To validate the effectiveness of the proposed method, we designed experiments. Specifically, for DeepJSCC and SCDA, we train a decoder to reconstruct the original image based on illegally obtained features. For our proposed method, we simulate an attacker who, after stealing the signal, uses the same generative model to produce visual data. By comparing the differences between the images reconstructed by the simulated attacker and the original image, we evaluate the privacy protection capability of each method.

¹<https://github.com/SSG2019/ULSC>

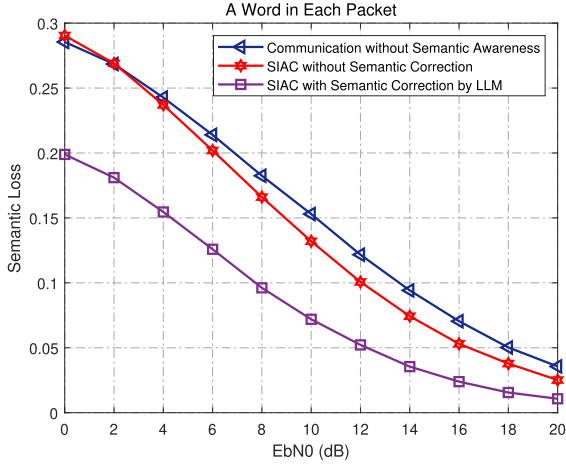


FIGURE 4. Semantic loss of different transmission strategies.

We use Peak Signal-to-Noise Ratio (PSNR) as a metric to quantify this difference, with the following expression:

$$Attacker-PSNR = 10 \log_{10} \frac{MAX^2}{MSE} \quad (dB), \quad (19)$$

where $MSE = d(\mathbf{S}, \hat{\mathbf{S}})$ is the mean squared-error between the original image $\mathbf{S}$ and the generated image $\hat{\mathbf{S}}$, and MAX is the maximum value of the image pixels. The images we used are 24-bit depth RGB images with each color channel having 8 bits per pixel, so $MAX = 2^8 - 1 = 255$. The smaller the PSNR, the greater the difference, and the stronger the privacy protection ability.

B. SUPERIORITY OF SEMANTIC-AWARE ADAPTIVE MODULATION AND CODING

The modulation scheme is selected from the set $\mathcal{M}_{all} = \{BPSK, QPSK, 16QAM, 64QAM, 256QAM\}$. The channel coding schemes is chosen from $\mathcal{C}_{all} = \{(2^k, k, 2^{k-1})\}$ Hadamard codes $|k = 3, 4, 5, 6\}$ with block length $B_l = 2^k$, code rate $R_l = k/2^k$ and minimum distance $d_{l,min} = 2^{k-1}$. It is worth noting that Hadamard codes are chosen as it has closed-form expressions of block length, coding rate, and minimum codeword distance. The proposed AMC can also be extended to selecting other MCS. The frame header is assumed to be 10 Bytes. In the experiment, we investigate two packeting solutions, one packing one word in a packet, and the other packing two words in a packet. The delay threshold is set to 1 ms and 2 ms for transmitting all packets for the above two cases, respectively. The MCS candidate set size N_{set} is set to 1000. The experiment results are illustrated in Figs. 4 and 5, respectively. It is showed that in both experiment setups, the semantic-aware communication scheme considerably outperforms conventional communication without semantic awareness. With the assistance of LLM at the receiver for semantic error correction, the semantic loss can be greatly reduced, reflecting the effectiveness of LLM for communication performance improvement. The performance gain in a word in each packet is more significant as in this

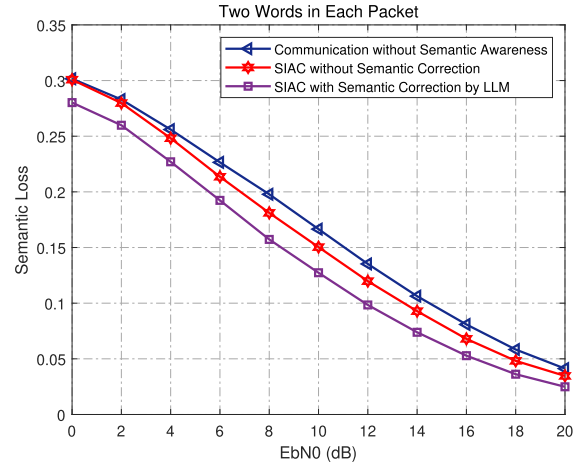


FIGURE 5. Semantic loss of different transmission strategies.

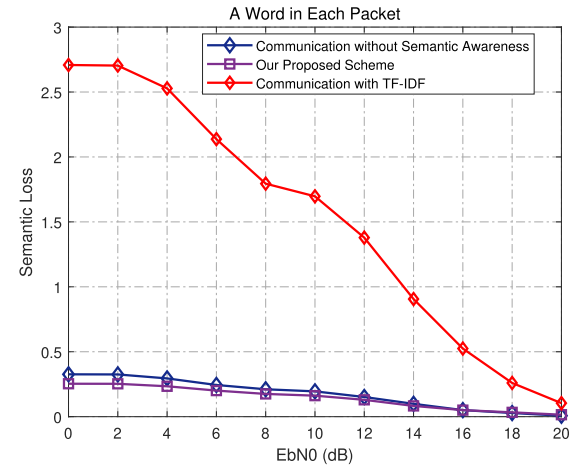


FIGURE 6. Semantic loss of different semantic awareness strategies.

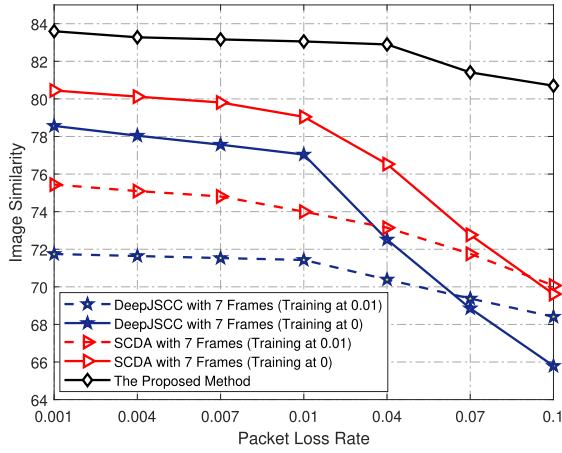
case, the LLM can handle one word missing compared to two words missing.

Next, we further verify the validity of the proposed strategy for quantifying semantic importance. Specifically, we choose to compare with the TF-IDF scheme. TF-IDF is a commonly used weighting technique for information retrieval and text mining. The TF-IDF helps to assess how important a word is to a text. Fig. 6 shows the semantic loss of the communication scheme without semantic awareness, the communication scheme with TF-IDF semantic awareness, and our proposed scheme. It can be seen that our proposed scheme achieves lower semantic loss, while TF-IDF achieves the highest semantic loss, because TF-IDF relies on word frequency to determine importance and fails to capture semantic information.

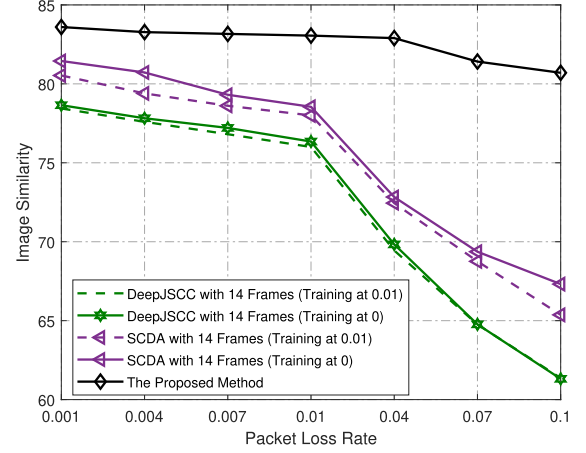
C. COMPARISON TO BASELINES OVER THE FRAME ERASE CHANNEL

1) PERFORMANCE COMPARISON

In this subsection, we evaluate the performance of the proposed method and the baselines at frame erase channel, that

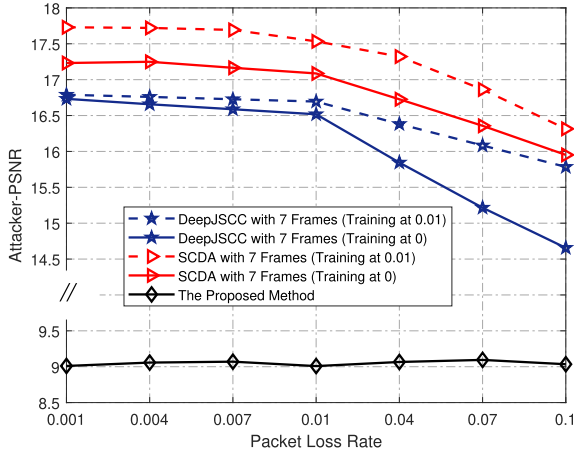


(a) Packaging all features into 7 frames

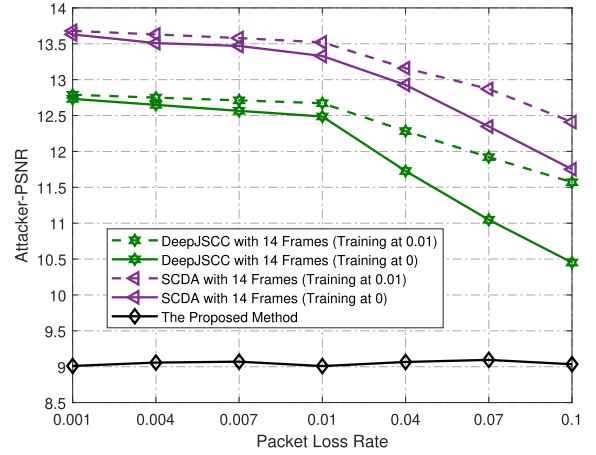


(b) Packaging all features into 14 frames

FIGURE 7. Image similarity of different methods versus frame erase rate.



(a) Packaging all features into 7 frames



(b) Packaging all features into 14 frames

FIGURE 8. Attacker-PSNR of different methods versus frame erase rate.

is, semantic similarities between transmitted and recovered semantic understanding. For all baselines, we set them to train at the frame erase rates of 0 and 0.01, respectively, and then test them at the frame erase rates of 0.001, 0.004, 0.007, 0.01, 0.04, 0.07, and 0.1.

Fig. 7 shows the image similarity of all methods versus frame erase rate. In Fig. 7(a), the transmitted features of baselines are packaged into 7 frames. It can be seen that there are significant differences in image similarity between different methods at different frame erasure rates. When the frame erasure rate is low (e.g. 0.001 to 0.01), the image similarity of DeepJSCC (Training at 0.01) remains around 0.72 and that of SCDA (Training at 0.01) remains around 0.76. As the frame erasure rate increases, the image similarity decreases slightly, but the overall image remains relatively stable. In contrast, at low frame erasure rates (e.g. 0.001 to 0.01), the image similarity of DeepJSCC (Training at 0) remains around 0.78 and that of SCDA (Training at 0) remains around 0.80, but their image similarity decreases significantly when the frame erase rate reaches 0.1, demonstrating sensitivity to high frame

erasure rates. In Fig. 7(b), the transmitted features of baselines are packaged into 14 frames. DeepJSCC and SCDA perform slightly better in most cases, however, with the increase of frame erase rate, their image similarity decreases rapidly, especially under the frame erase rate of 0.1. In contrast, our proposed method exhibits the best image similarity at all frame erase rates. On a scale of 0.001 to 0.1, image similarity remains between 0.82 and 0.84. Even at a high frame erase rate of 0.1, the image similarity of our proposed method is much higher than that of all baselines. On the whole, our proposed method is significantly better than the baselines in each frame erase rate, showing better robustness and stability.

2) VALIDATION OF VISUAL PRIVACY PROTECTION CAPABILITIES

To verify the advantages of the proposed method in protecting privacy, we conduct model inversion attacks on the proposed method and all baseline methods. In Fig. 8, we show the Attacker-PSNR versus frame erase rate. We can



FIGURE 9. From left to right are the original image, and the images obtained by model inversion attacks on DeepJSCC with 7 Frames (Training at 0.01), DeepJSCC with 7 Frames (Training at 0), SCDA with 7 Frames (Training at 0.01), SCDA with 7 Frames (Training at 0), and the proposed method. For each reconstructed image in evaluation, the frame erase rate is set to 0.001.

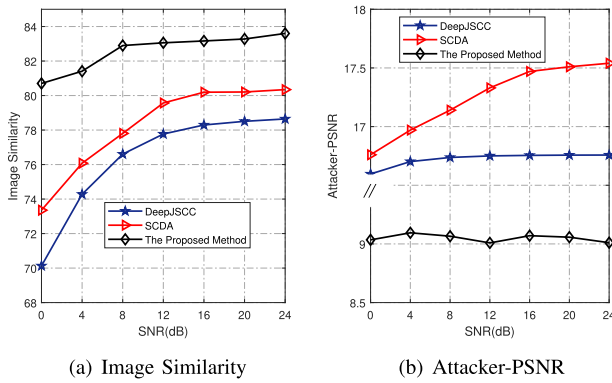


FIGURE 10. Performance comparison versus SNR on the AWGN channel.

observe that different methods have significant differences in Attacker-PSNR under different frame erase rates. Specifically, in Fig. 8(a), for DeepJSCC and SCDA trained at a frame erase rate of 0.01, at low frame erase rates (e.g. 0.001 to 0.01), DeepJSCC's Attacker-PSNR stays around 16.8 and SCDA's Attacker-PSNR stays around 17.8, and they change little. As the frame erase rate increases, their Attacker-PSNR slightly decreases. For DeepJSCC and SCDA trained at a 0 frame erase rate, the Attacker-PSNR drops faster as the frame erase rate increases. In contrast, our proposed method has significantly lower Attacker-PSNR at all frame erase rates than the baseline methods, with Attacker-PSNR remaining between 8.5 and 9 in the range of 0.001 to 0.1. This shows that our proposed method makes it difficult for attackers to obtain high-quality images regardless of how frame erase rates vary. Fig. 8(b) shows the same tendency.

To more intuitively demonstrate the privacy benefits of our proposed approach, we present images obtained by adversaries using model inversion, as shown in Fig. 9. We can see that the image obtained by attacking the baseline method is the same person as the original image, although there is some

blur and noise. Although the image obtained by attacking the proposed method is clear, it can be clearly seen that it is different from the person in the original image. This effectively protects the user's visual privacy, indicating the advantages of the proposed method.

D. COMPARISON TO BASELINES OVER THE AWGN CHANNEL

To more fully demonstrate the robustness and adaptability of our proposed system, we also conduct tests on AWGN channel. The test results are shown in Fig. 10. Fig. 10(a) shows the performance of DeepJSCC, SCDA and our proposed method in terms of Image Similarity under different SNR regimes. It can be seen from the figure that the proposed method obtains the highest image similarity value under both low and high SNR regimes, which indicates that the proposed method can reliably transmit semantics even under noise interference environments. Fig. 10(b) shows the performance of different methods in terms of visual privacy protection under the same conditions. Our proposed method achieves the lowest Attacker-PSNR in all SNR tested, which indicates that the method has significant advantages in protecting users' visual privacy.

VII. DISCUSSION

By introducing LLMs, our proposed ULSC system makes significant progress in guaranteeing semantic transmission and enhancing visual privacy. Although the current research results are promising, there are still some key areas that deserve further exploration and optimization to further improve the performance and utility of the system.

A. SEMANTIC IMPORTANCE EVALUATION

In our current approach to semantic importance evaluation, we have primarily focused on individual word assessments. While this method has proven effective for many

applications, it falls short in capturing the contextual significance of phrases or word combinations. For example, in the sentence “A beach with palm trees and clear blue water”, the individual words “palm” and “trees” may be undervalued due to their common co-occurrence. However, when considered as the phrase “palm trees”, they form a critical component of the scene’s description. This discrepancy reveals a key limitation in our method: the potential omission of important semantic relationships between words.

To overcome this limitation, we plan to explore semantic importance evaluation method based on image semantic segmentation [41]. For example, we consider an image that the text describes as “a beach with palm trees and clear blue water”. Palm trees and other important elements in the images can be identified by semantic segmentation. If the segmentation results show that “palm trees” occupies a prominent position in the vision, then the system will improve the semantic importance of the phrase “palm trees” in the overall text description. This method not only captures keywords in the text, but also understands how these keywords combine with other visual elements to form a complete description of the scene. By integrating text and image, the semantic importance of each part of text can be determined more accurately, so as to improve the performance and effect of semantic communication system.

B. SEMANTIC ERROR CORRECTION WITHOUT PRIOR KNOWLEDGE OF ERROR LOCATIONS

In Section V, we introduced a BERT-based semantic error correction method that relies on the receiver knowing exactly where the data is lost. However, this assumption often fails in practical applications. To solve this problem, a new error correction mechanism can be introduced that does not require prior knowledge of the error location.

Sequence number [42] and checksum [43] are two widely used techniques that can help identify missing or corrupted data frames during data transmission. Sequence number ensures sequential data by assigning a unique identifier to each packet. If the receiver detects a discontinuity in the sequence number, it can determine that the data frame has been lost. Checksum is hashed over the packet to generate a short checksum, which is then appended to the data frame. The receiver verifies the integrity of the data by recalculating the checksum of the received data frame and comparing it with the transmitted checksum.

The new mechanism first uses sequence number to detect missing data frames. Once a data frame loss is detected, the system will enter error correction mode. In error correction mode, checksum is used to determine if the packet is corrupt. If the data frame is corrupted, the system will attempt to recover the lost data using a BERT-based semantic error correction method. By introducing sequence number and checksum, the system can perform efficient semantic error correction without knowing the location of the error. This improvement is expected to improve the practicability and

robustness of the designed system and make it more suitable for practical communication environment.

C. MODEL CUSTOMIZATION AND DEVELOPMENT

In this paper, we utilize pre-trained models such as BERT and ChatGPT due to their established effectiveness in natural language processing tasks. These models provide robust semantic understanding, which enhances the accuracy of our analysis and allows for efficient development [44]. Leveraging these mature technologies enables us to focus on the core innovations of our semantic communication framework.

We recognize the importance of customizing these pre-trained models to better meet the specific needs of semantic communication tasks. Our future work will involve modifying model architectures and training processes to optimize the conversion of visual data into natural language descriptions. Additionally, we will design task-specific loss functions and evaluation metrics to further refine model performance. We also plan to fine-tune the models using relevant datasets and explore transfer learning strategies to enhance their adaptability to particular communication environments. These efforts aim to strengthen the overall effectiveness of our proposed system in addressing the challenges inherent in semantic communication.

VIII. CONCLUSION

In this study, we proposed an innovative approach for understanding-level semantic communications by integrating semantic importance-aware communication (SIAC) with semantic correction using advanced LLMs such as BERT and ChatGPT. Our method aligns image communication within the human-comprehensible semantic space of natural language, enabling effective semantic correction and significance analysis through LLMs. We developed a technique to quantify the semantic value of data in each frame, supporting a semantically informed SIAC strategy. Additionally, the research explores a semantic-aware adaptive modulation and coding scheme, complemented by a proposed low-complexity greedy search method to ensure semantically robust communications within set latency limits. The utilization of LLMs for filling in missing frames is also introduced. Experimental results validate the superiority of our approach in enhancing agent-human communications or generative task-oriented communications, particularly in terms of minimizing semantic loss and bolstering privacy protection.

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